

Roll No.....

Mid Semester Examination, September 2025

SET-I

Name of the Program: **B. Tech. Biotechnology**

Semester: *V*

Name of the Course: **Bioinformatics**

Course Code: TBT504

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1		(10 marks)	
(a)	What is INSDC? Discuss the databases that are part of INSDC. How is data redundancy avoided while including data in the databases.		CO1
	OR		
(b)	OMIM is a database used to find information about human genetic disorders. What type of biological database it is? Discuss the other types of biological databases with examples.		
Q2		(10 marks)	
(a)	Explain the fundamental principles and operational mechanism of BLAST in the context of sequence alignment. Analyze the different BLAST variants and highlight their specific applications in biological research.		CO4
	OR		
(b)	Using the Needleman-Wunsch algorithm, determine the optimum global alignment for the sequences S1 = GATTACA and S2 = GCATGCU given the scoring scheme: Match = +1, Mismatch = -1, Gap penalty = -2. Construct the scoring matrix, perform the traceback procedure, and thoroughly justify your alignment outcome.		
Q3		(10 marks)	
(a)	Explain the role of scoring matrices in sequence alignment. Discuss the differences between PAM and BLOSUM matrices and evaluate their applications in DNA versus protein sequence analysis.		CO3
	OR		
(b)	Assess the importance of Multiple Sequence Alignment (MSA) in biological investigations. Describe and compare progressive and iterative alignment methods, highlighting their strengths and drawbacks.		
Q4		(10 marks)	
(a)	Explain the concept of medical and genome databases with relevant examples. Discuss their roles in facilitating our understanding of human diseases, drug discovery, and precision medicine.		CO2
	OR		
(b)	What are molecular fossils? Discuss their role in phylogeny. Discuss the modern day steps used in phylogenetic analysis.		
Q5		(10 marks)	
(a)	A researcher wants to predict gene function and evolutionary relationships using database resources. Which databases and bioinformatics methods should be used for functional annotation and phylogenetic analysis? Discuss the workflow and rationale.		CO4
	OR		
(b)	Describe the challenges involved in storing and retrieving large-scale biological data. How do bioinformatics databases such as NCBI, EMBL, and DDBJ address these challenges and ensure accessibility and reliability for researchers?		

